

Krishna Gaur

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EDUCATION

Indian Institute of Information Technology, Bhopal

Bachelor of Technology in Computer Science and Engineering (CSE)

CGPA: 8.67 | Sep 2023 – Jun 2027

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Software Engineering, Operating Systems, Computer Networks, Distributed System

PROJECTS

ADAPTIVE TRAFFIC SIGNAL OPTIMIZATION | *Spring Boot, Java, Python, SUMO, TraCI*

[GitHub](#)

- Engineered a real-time adaptive traffic signal control system that dynamically optimized 4-phase intersection timings using live congestion deviation analysis
- Implemented adaptive cycle expansion, normalization, and constraint-based green-time redistribution algorithms for congestion-aware signal scheduling under varying traffic loads
- Architected a modular Java–Python simulation pipeline integrating SUMO and TraCI to enable scalable experimentation, parameter tuning, and reproducible benchmarking
- Simulated 400+ vehicle movements per experiment across diverse traffic scenarios including peak-hour, gridlock, balanced, and highly unbalanced traffic distributions
- Achieved **2.4% average reduction** in vehicle waiting time over Fixed-Time Signal (FTS) systems, with up to **4.4% improvement** in extreme congestion imbalance scenarios, evaluated using Average Waiting Time (AWT)

SPENDWISE - Expense Tracker Web App | *Spring Boot, React, Java, PostgreSQL, JavaScript, Tailwind*

[GitHub](#)

- Designed a secure multi-user expense tracking platform with JWT authentication and email verification
- Performed unit and integration testing using JUnit, Mockito, MockMvc, covering critical API flows
- Built 15+ RESTful APIs using Spring Boot and JPA to manage expenses and income records
- Improved API response time by ~30% through query optimization and efficient data handling
- Ensured robustness by validating inputs and handling edge cases such as empty datasets and invalid ranges

DATA STRUCTURES & ALGORITHMS

- Solved **500+** data structures and algorithmic problems across LeetCode, Codeforces, CodeChef, and GeeksforGeeks
- [Codeforces](#): Pupil (Max rating: 1537)
- [CodeChef](#): 4 Star (Max rating: 1805)

HACKATHONS & LEADERSHIP

Smart India Hackathon 2024 – Team Leader & Full Stack Developer

- Led backend development for a team of 6 under strict deadlines and rapid iteration cycles and decision making
- Developed RESTful backend APIs to support real-time client–server communication and synchronized data exchange
- Optimized development workflows and task coordination, improving team execution efficiency by ~20%

SKILLS

- Programming:** C++, Java, TypeScript, JavaScript, HTML, SQL
- Backend:** Spring Boot, Maven, REST APIs
- Frontend:** React, Tailwind CSS
- Databases:** MongoDB, PostgreSQL, MySQL, H2 (testing)
- DevOps & Tools:** Docker, Git, GitHub, CI/CD, Cloud Deployment